STATSEI14 · PRACTICAL TUTORIAL

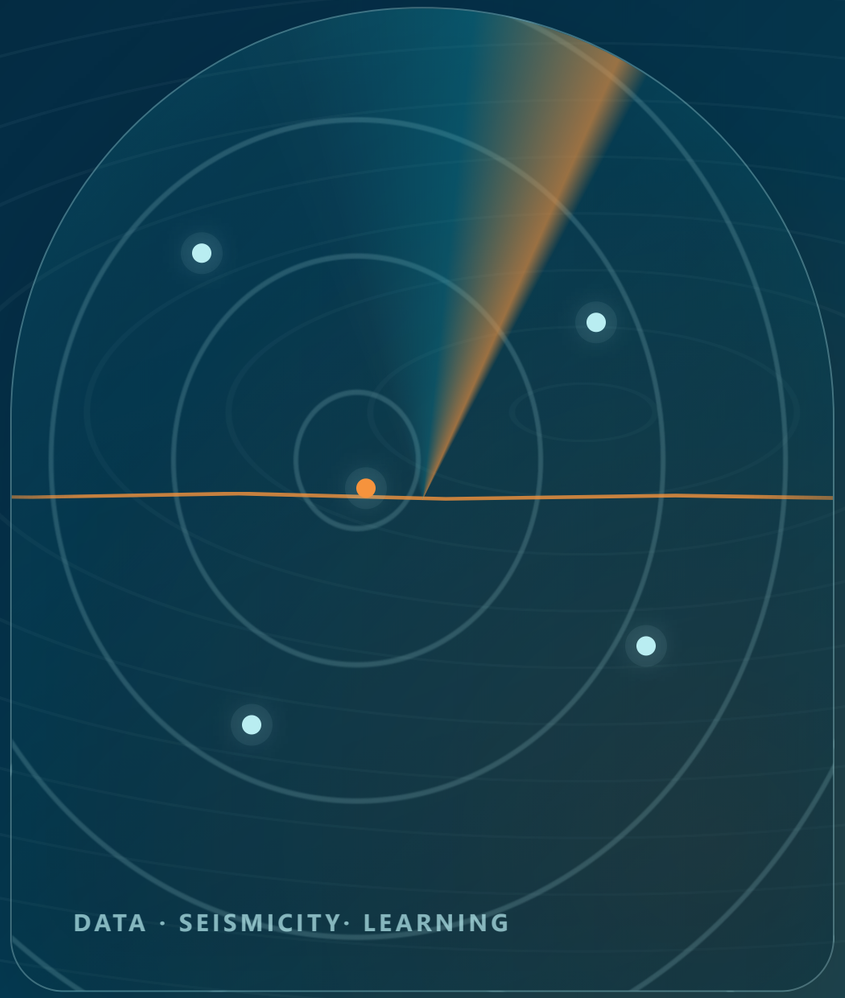
Deep Learning Meets ETAS

Practical Hybrid Workflows for Seismic Forecasting

Francisco Plaza-Vega
Universidad de Santiago de Chile

13 October 2026 · Chile

> How do we turn a seismic question into a deep-learning experiment?



From seismic questions to a learning experiment

Section	Guiding question
1. Seismic questions and data	What do we observe, and what do we want to estimate?
2. From data to learning tasks	How do we define inputs, targets and representations?
3. Deep-learning building blocks	How do different models use the available information?
4. Lessons from our research	What worked, what was difficult, and what did we learn?
5. Guided practical	How do we build and evaluate a weekly occurrence forecast?

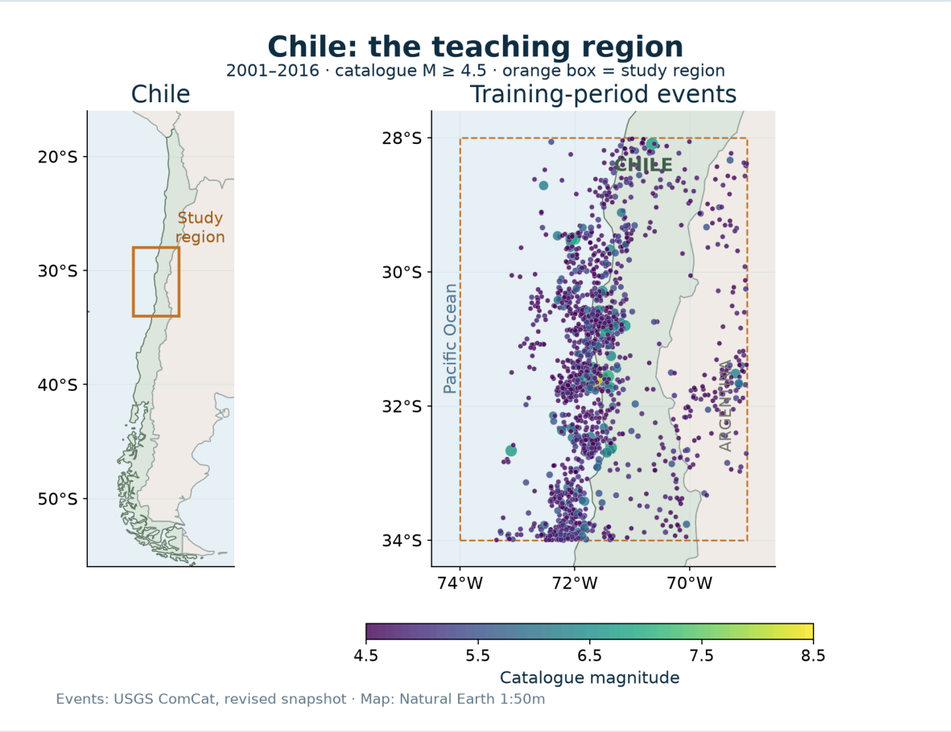
> How do we turn a seismic question into a deep-learning experiment?

01

Seismic questions and data

What do we observe, and what do we want to estimate?

What can we learn from seismicity?



Signals record ground motion.

Catalogues describe event times, locations and magnitudes.

Histories and maps organize information across time and space.

> The same catalogue can support different tasks. First define what the output should mean.

Map: frozen USGS training catalogue and Natural Earth. Illustrates today's regional setting.

Deep learning in seismology: inputs and targets

Example	Input	Learning task
ConvNetQuake · 2018	Recorded waveforms	Detect events and classify their source region
RECAST · 2023	Past event times and magnitudes	Estimate the next-event time distribution
Delgado et al. · 2025	Features of catalogued events	Classify foreshock, mainshock and aftershock labels
Farfán et al. · 2025	Sequences of daily spatial grids	Forecast gridded seismic activity

> Today’s practical: 30 days of history and a seven-day occurrence probability.

Perol et al. · Dascher-Cousineau et al. · Delgado et al. · Farfán et al.

02

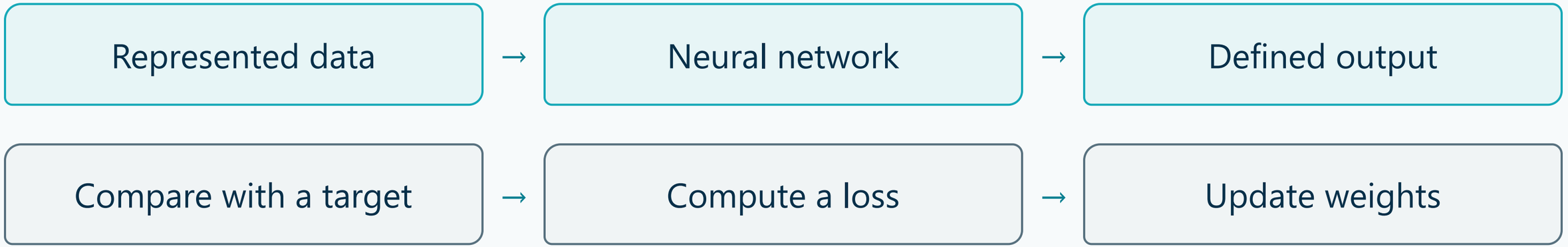
From data to learning tasks

How do we define inputs, targets and representations?

From a scientific question to a learning task

We define: the question, information available, input and target.

We represent inputs: as a vector, sequence or map.



Architecture combines information; **training** adjusts weights.

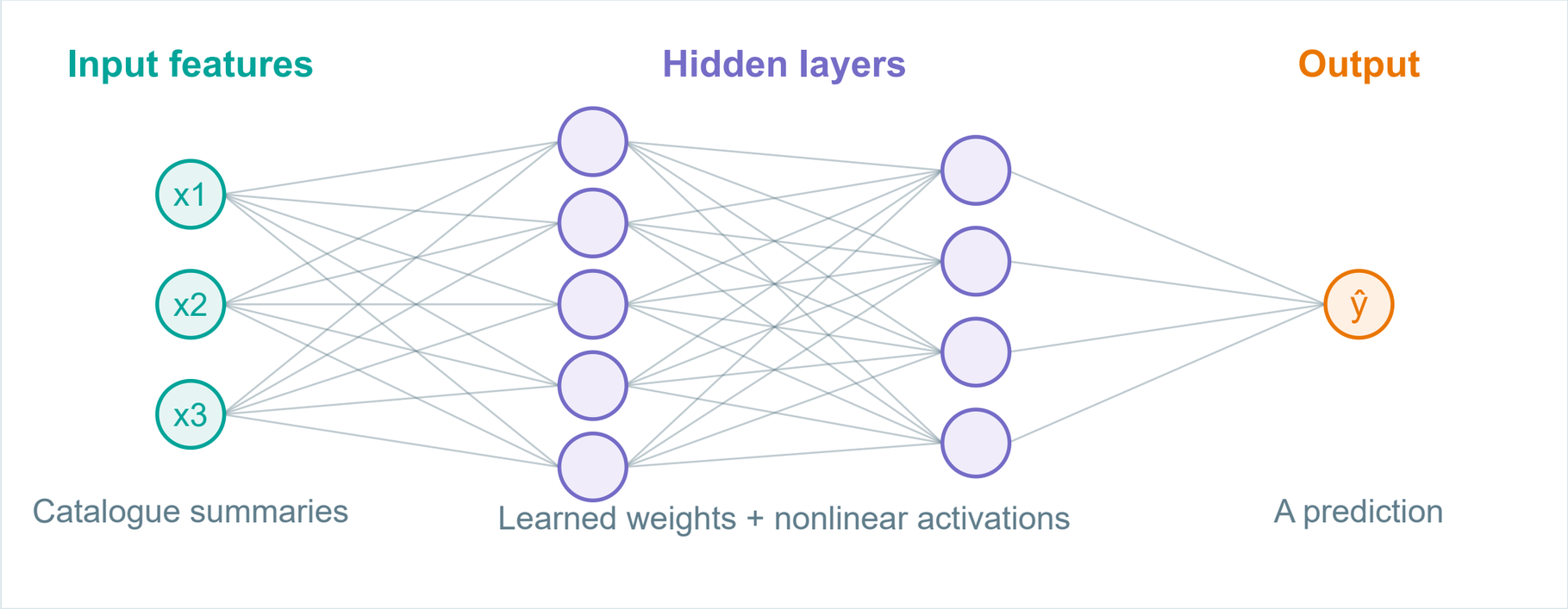
> The output and the loss must match the scientific question.

03

Deep-learning building blocks

How do different models use the available information?

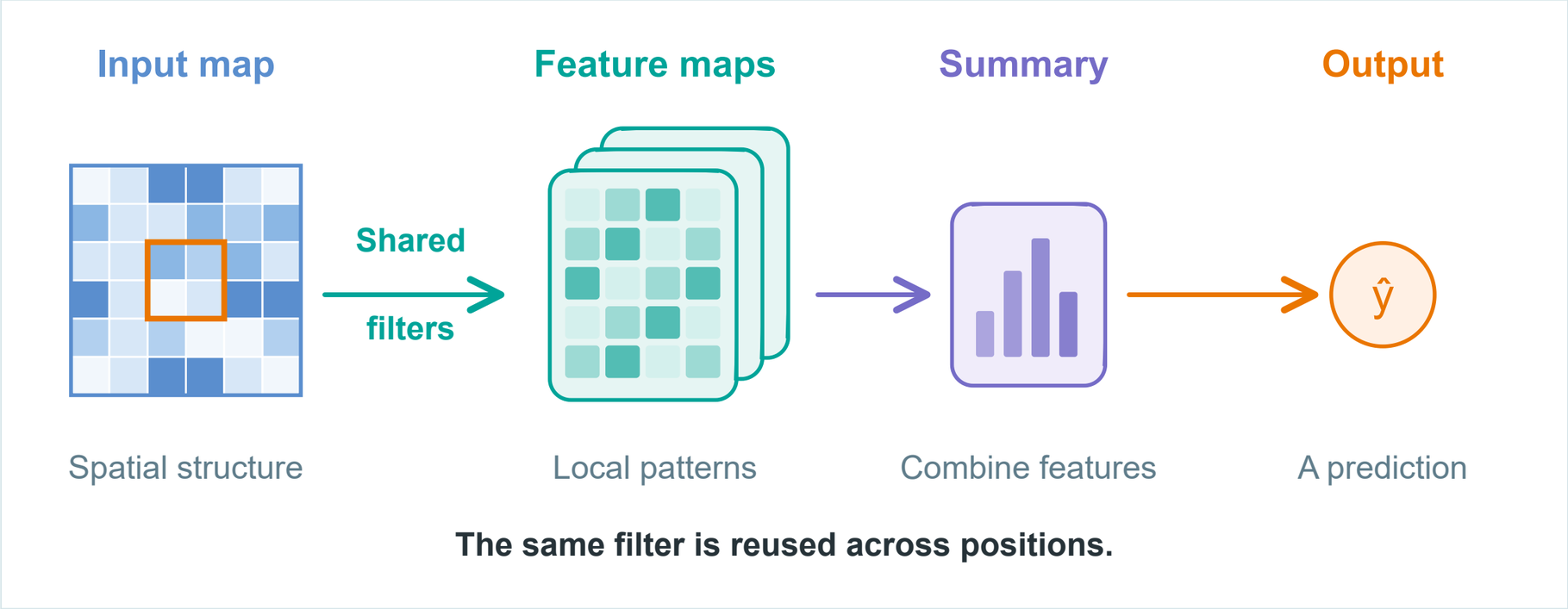
MLP: combine measured features



A multilayer perceptron learns nonlinear combinations of a feature vector.

Optional extension: the MLP uses 90 values at fixed lag positions. Diagram adapted from the author's DL course, Unit 2.

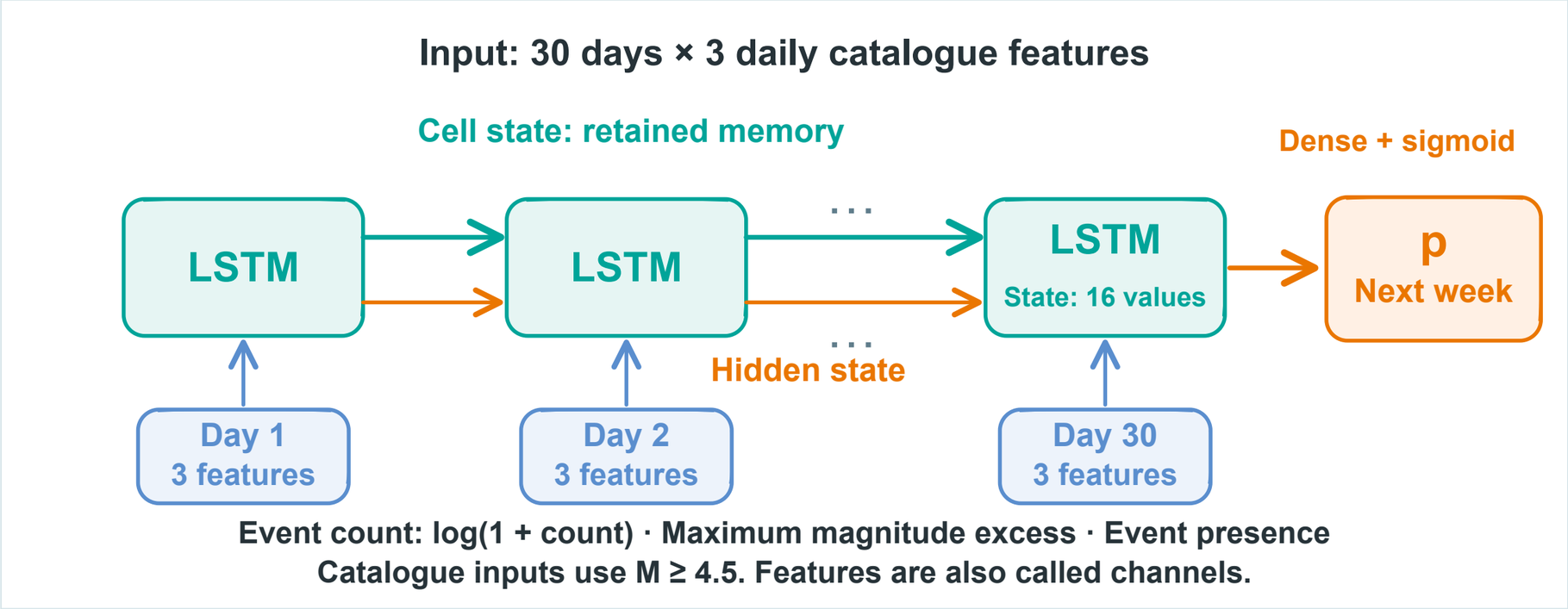
CNN: learn local patterns with shared filters



The same filter examines different neighborhoods.

Optional extension: CNN1D shares filters across days. Diagram adapted from the author's DL course, Unit 3.

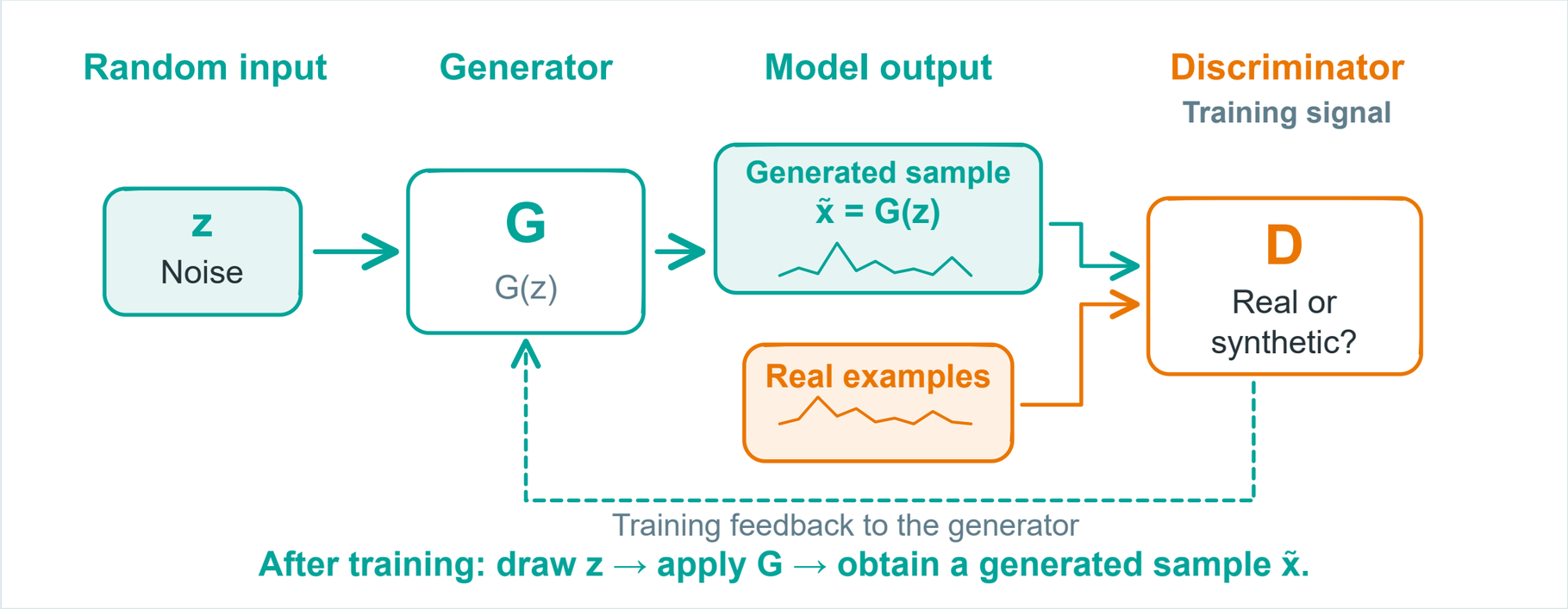
LSTM: carry information through a sequence



Today's example: 30 × 3 input → state (16) → Dense + sigmoid → weekly probability

Output: probability of at least one catalogue M ≥ 5 event in the next seven days.
Adapted from the author's DL course, Unit 4.

Generative learning: produce new samples



The output of G is a new data sample, such as a synthetic sequence.

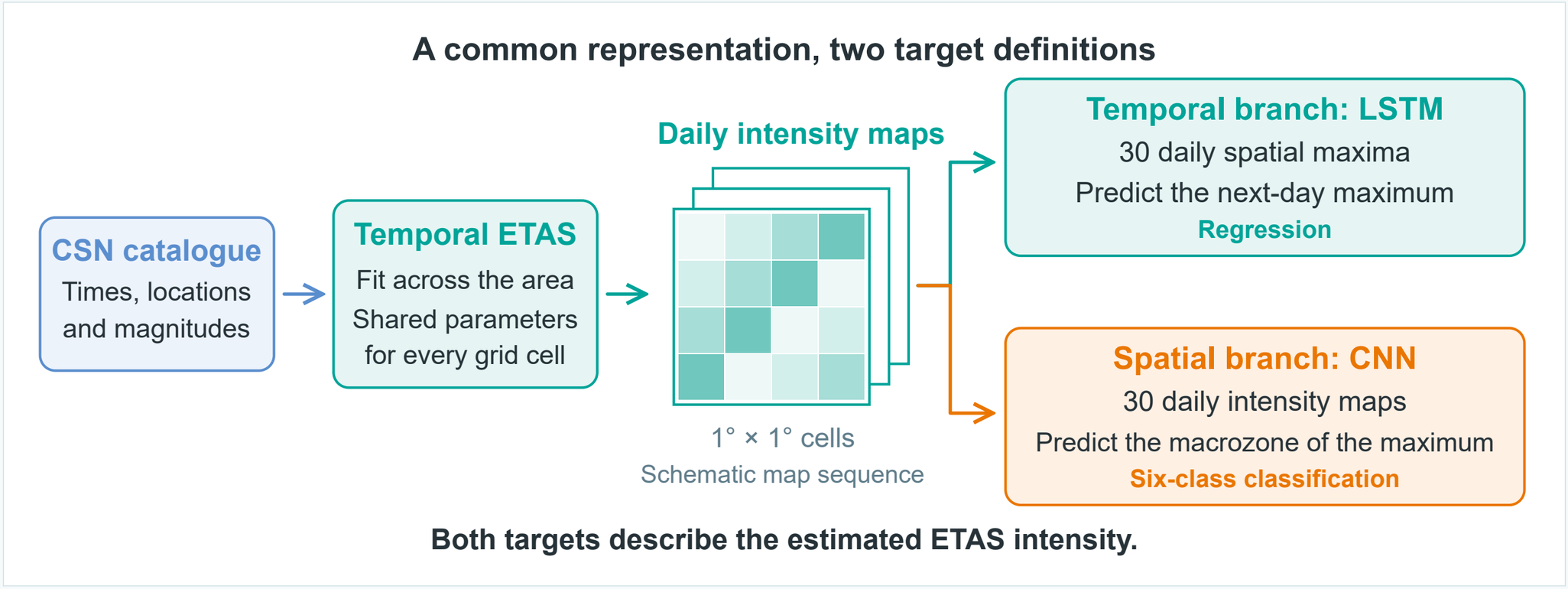
GAN is an adversarial learning scheme; G and D may use different neural architectures.
Later case: mainshock conditions + noise → a possible aftershock sequence.
GAN schematic adapted from the author's DL course, Unit 5.

04

Lessons from our research

What worked, what was difficult, and what did we learn?

ETAS intensity maps support two learning tasks



Nicolis, Plaza & Salas · Spatial Statistics 42 (2021), 100442 · CSN Chile, 2000–2017

ETAS intensity describes a conditional event rate

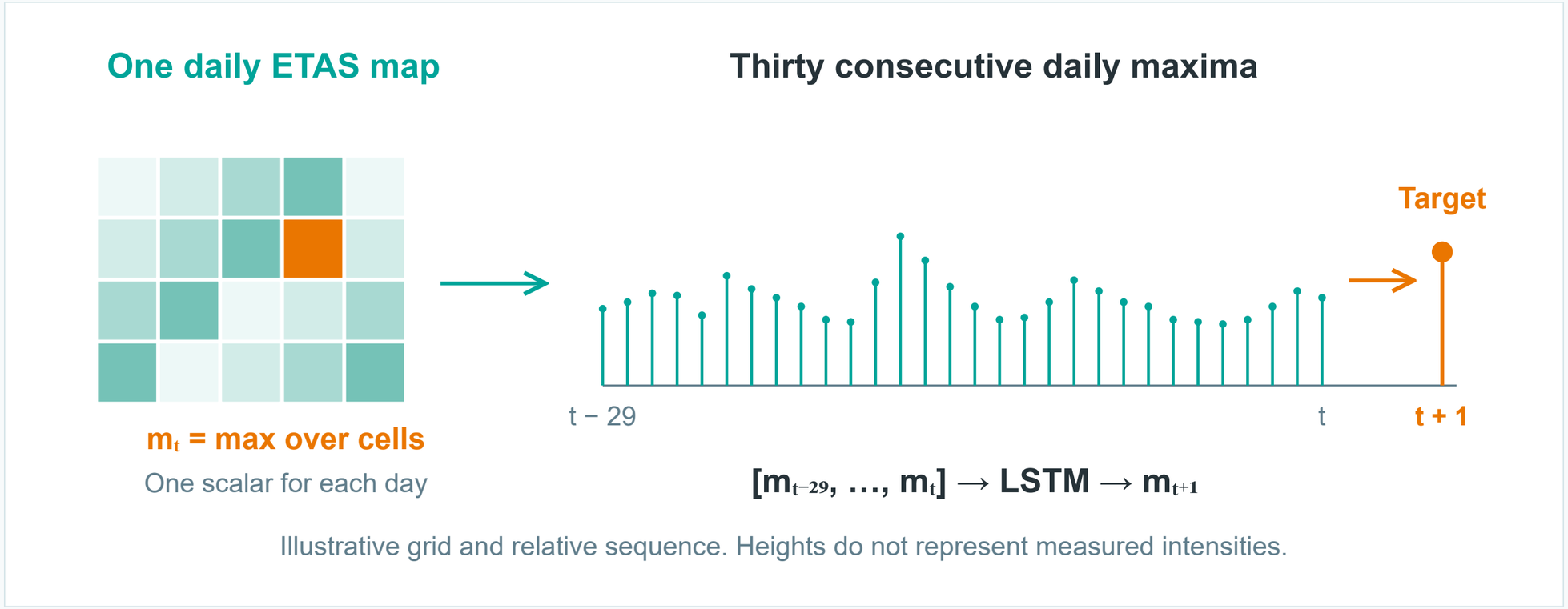
$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \text{contribution from event } i$$

Background + magnitude-dependent contributions that decay with time

Published construction: common ETAS parameters applied across 1° × 1° cells produce a sequence of intensity maps.

> A map value is an estimated occurrence rate, not an earthquake magnitude.

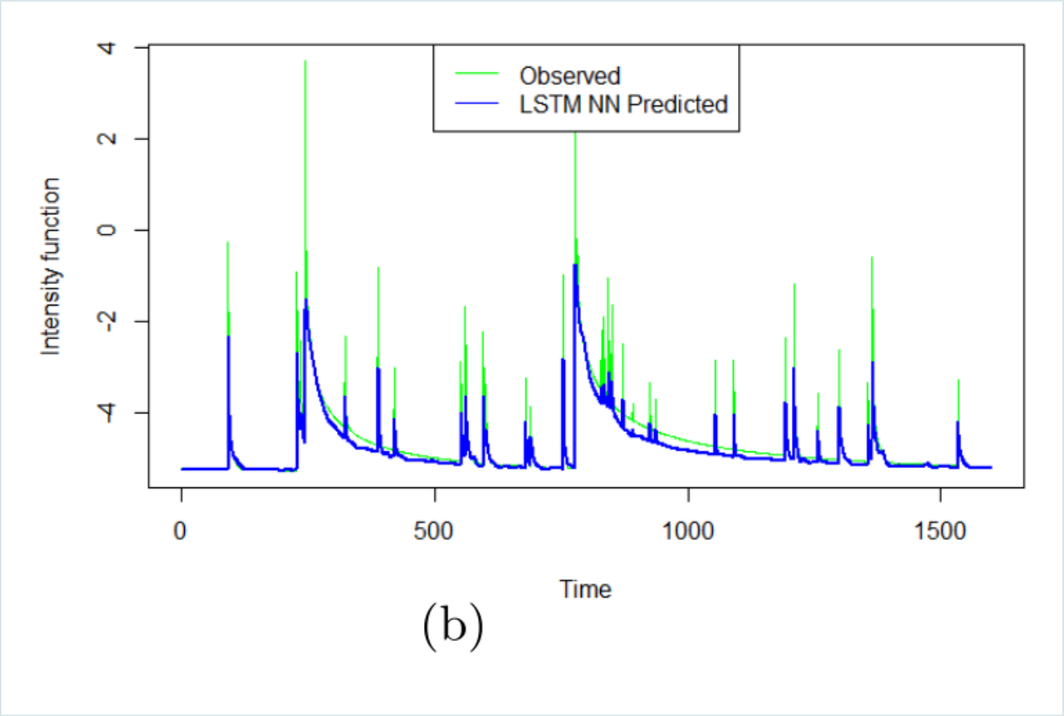
A map maximum becomes one value in the LSTM input



30 daily spatial maxima → LSTM → next-day maximum intensity

Nicolis, Plaza & Salas (2021) · Original published regression task · Diagram illustrates the operation, not measured values

Published LSTM result: an ETAS-derived test target



M ≥ 6 setting · Table 3

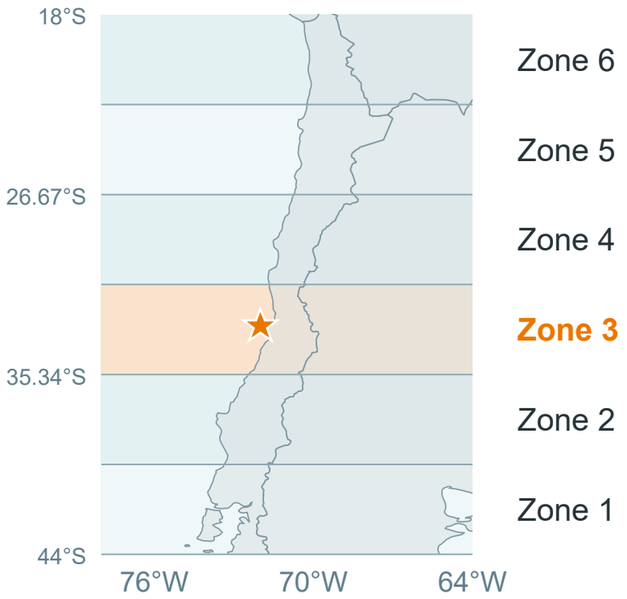
Model	RMSE	R^2
ETAS	0.5475	0.6241
LSTM	0.5296	0.6561

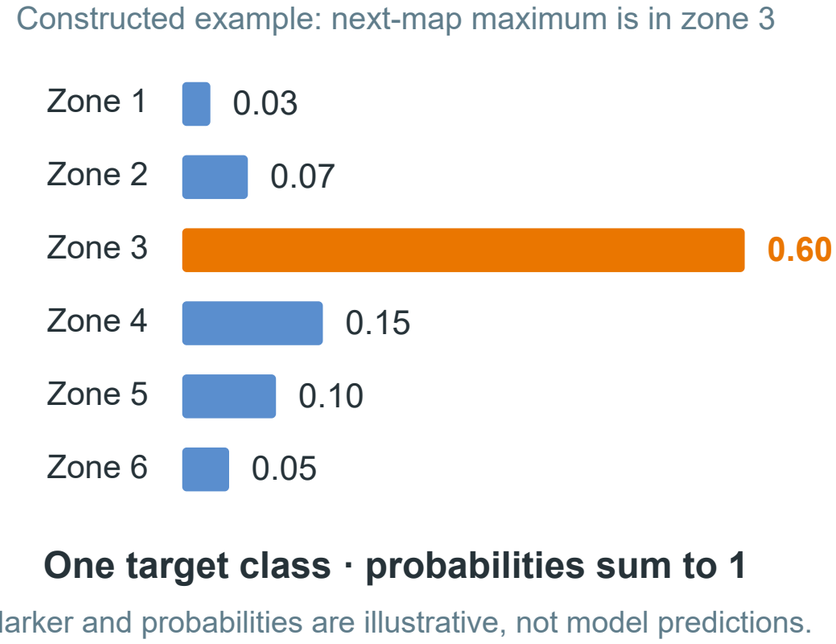
Lower error and higher R^2 for this reported comparison.

> The network tracks a model-derived target; this is not a direct event-probability test.

Nicolis, Plaza & Salas (2021), Figure 5(b) and Table 3 · Original plotting scale retained

CNN: locate the next map maximum in one of six macrozones





The target is one macrozone containing the next day's maximum ETAS intensity.

Nicolis, Plaza & Salas (2021), Table 2 · Natural Earth geography · Marker and probabilities are illustrative

Generative exploration of aftershock sequences

Mainshock conditions

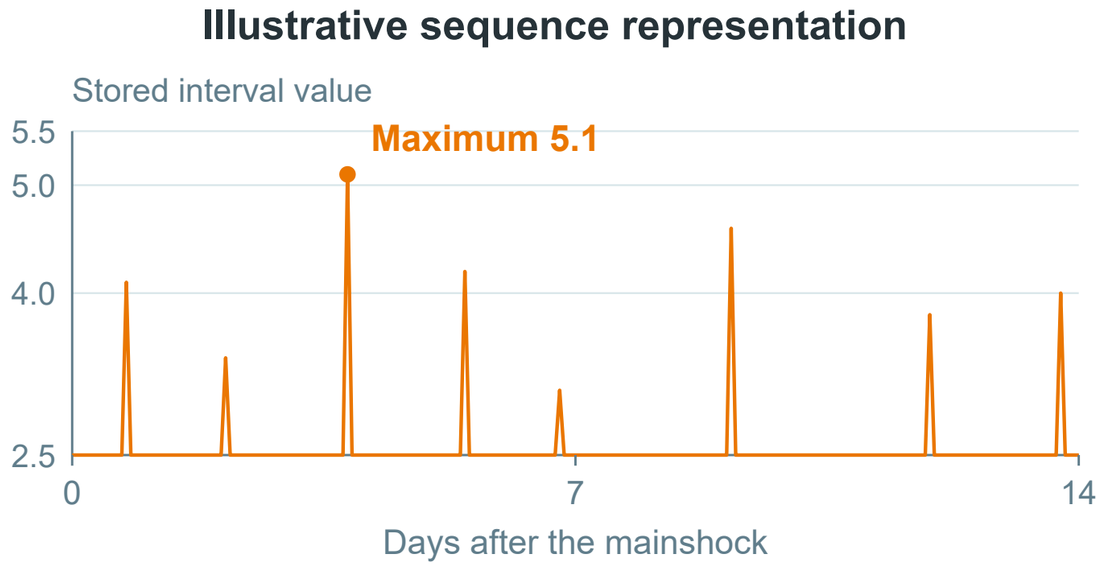
Location, magnitude, depth

+ random noise

↓

Conditional generator →

cGAN / WGAN-GP / TGAN



224 intervals of 90 minutes. The value 2.5 marks absence.

Values chosen for teaching; no empirical forecast is shown.

The intended output is a possible sequence conditioned on a mainshock.

Jara Muñoz (2025) · Undergraduate thesis in Statistical Engineering · Universidad de Santiago de Chile

An observed and generated sequence: one manuscript case

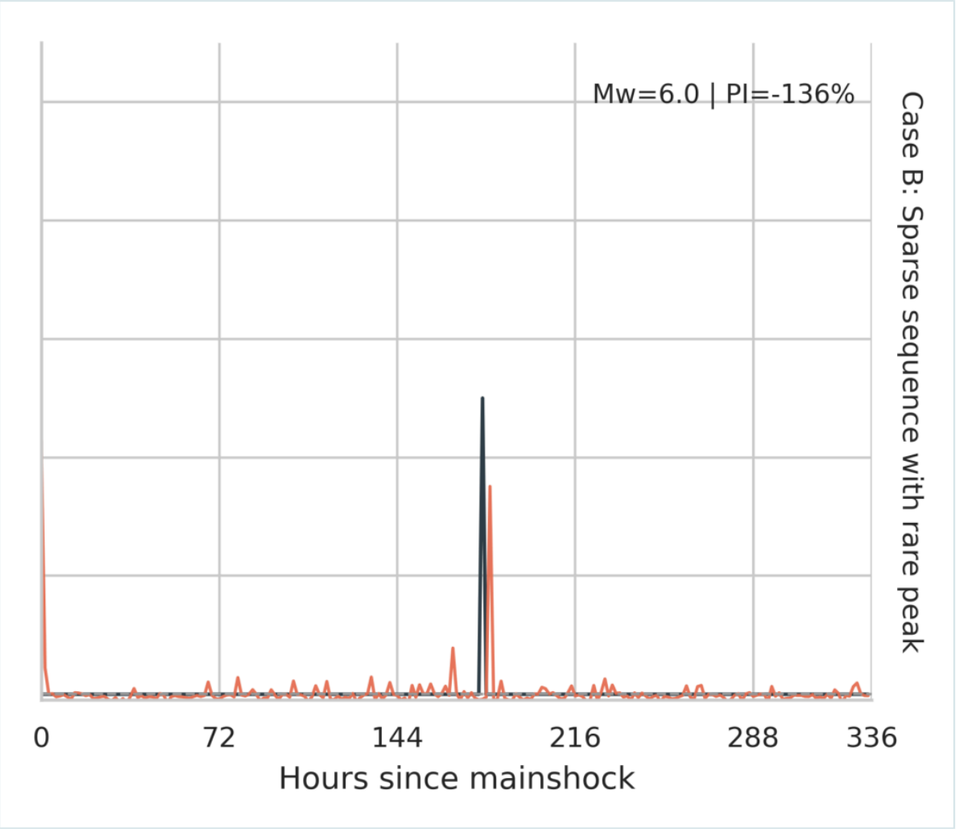
GANT · case B

—— Observed reference

—— Generated sequence

Vertical axis: interval maximum magnitude (Mw).

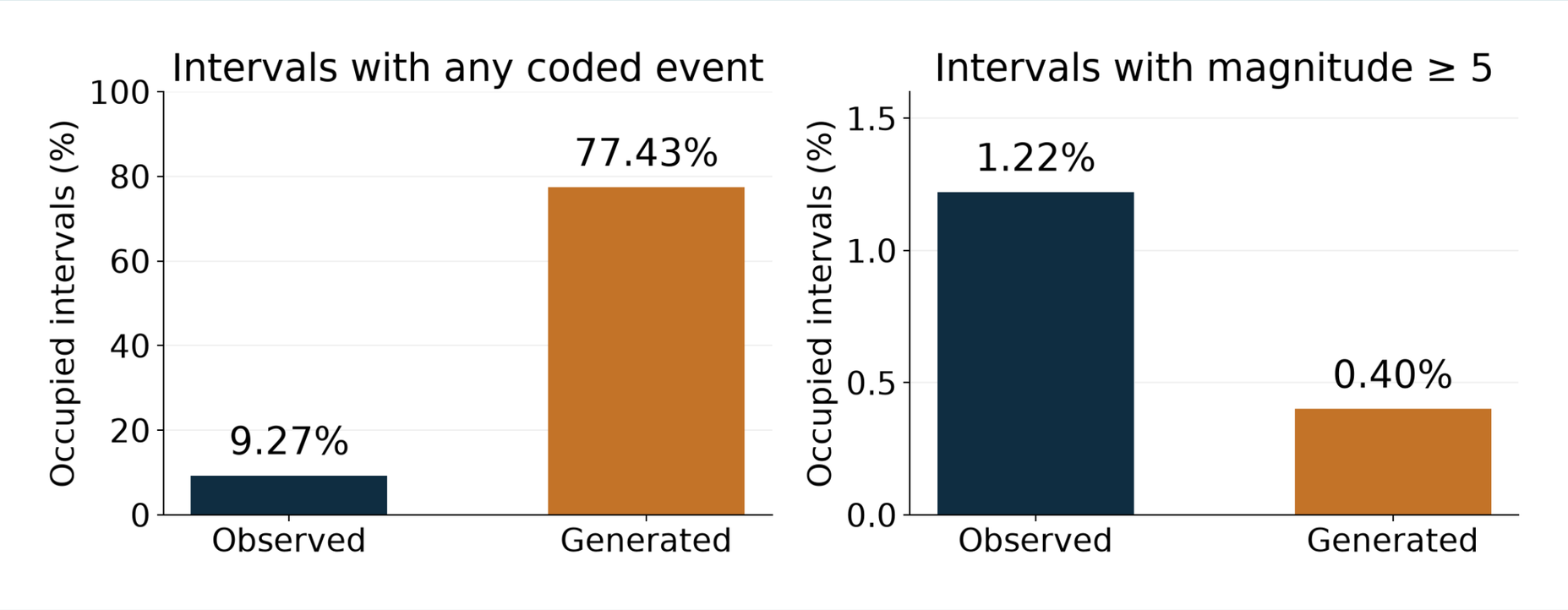
2.5 marks absence.



> Compare occurrence, peak magnitude and timing, as well as the shape of the curve.

2026 manuscript under review, Figure 4 · One case from the primary random split · Original PI = −136% for this case

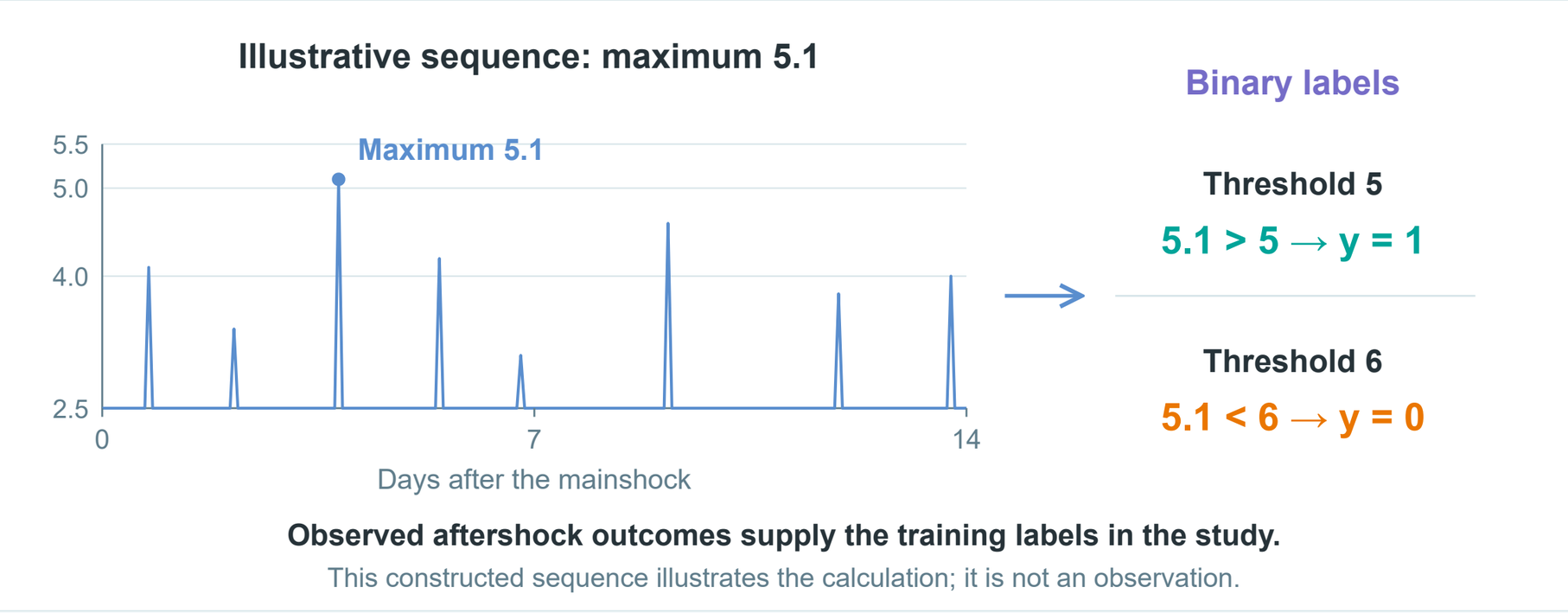
GANT temporal evaluation: too many occupied intervals



GANT overproduces occupied intervals while underrepresenting intervals with M ≥ 5.

2026 manuscript under review · Temporal evaluation · Percentages of intervals, not total event counts

Observed sequences define magnitude-threshold labels



Observed aftershock outcomes supply the training labels in the study.

This constructed sequence illustrates the calculation; it is not an observation.

> Mainshock features → classifier → probability of at least one qualifying aftershock

Manuscript under review · 14 days, 100 km · Observed outcomes define labels; the displayed sequence is constructed

Classification performance depends on the evaluation

Reported target (Table 9): at least one Mw ≥ 6 aftershock within 14 days and 100 km.

Evaluation scheme	Best reported classifier	ROC-AUC
Random split	Random Forest	0.816
Temporal split	Gradient Boosting	0.672
Geographic split	Random Forest	0.642

Each row selects the best model in its own evaluation. Models and evaluation samples differ.

> These AUCs measure discrimination. They do not establish calibrated probabilities.

Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review · Table 9 · Magnitude-threshold classification

Research hypothesis: ETAS-informed generation

History available at the origin → ETAS-derived information → Conditional generator

Possible role: supply a conditioning variable or a constraint on generated sequences.

Question: does that addition improve occurrence, magnitude and temporal structure under a fixed evaluation?

> This is a proposed research direction. No hybrid results are shown here.

Future work discussed in the manuscript · The tutorial's illustrative activity score is a separate feature experiment

05

Guided practical

How do we build and evaluate a weekly occurrence forecast?

The practical applies the same modelling decisions

Shared decisions: question → representation → model and output → evidence.

Case	Representation	Output
Spatial Statistics	ETAS maps and daily maxima	Intensity maximum and its macrozone
Generative study	Mainshock conditions and noise	A possible binned aftershock sequence
Classification study	Mainshock predictors; observed outcome labels	Occurrence or exceedance probability
Today's practical	Daily catalogue channels	Weekly occurrence probability

> Now: 30 days of catalogue history → probability of at least one M ≥ 5 event in the next seven days.

Core: references, LSTM and probability evaluation. Optional: Transformer and spatial comparisons.

From the catalogue to three daily features

For each day, summarize catalogue events with M ≥ 4.5 inside the study region.

Daily feature	Definition	What it describes
Event count	$\log(1 + N_t)$	How many input events occurred
Maximum magnitude excess	Maximum magnitude − 4.5; zero if empty	How far the maximum exceeds 4.5
Event presence	1 if $N_t > 0$; otherwise 0	Whether any input event occurred

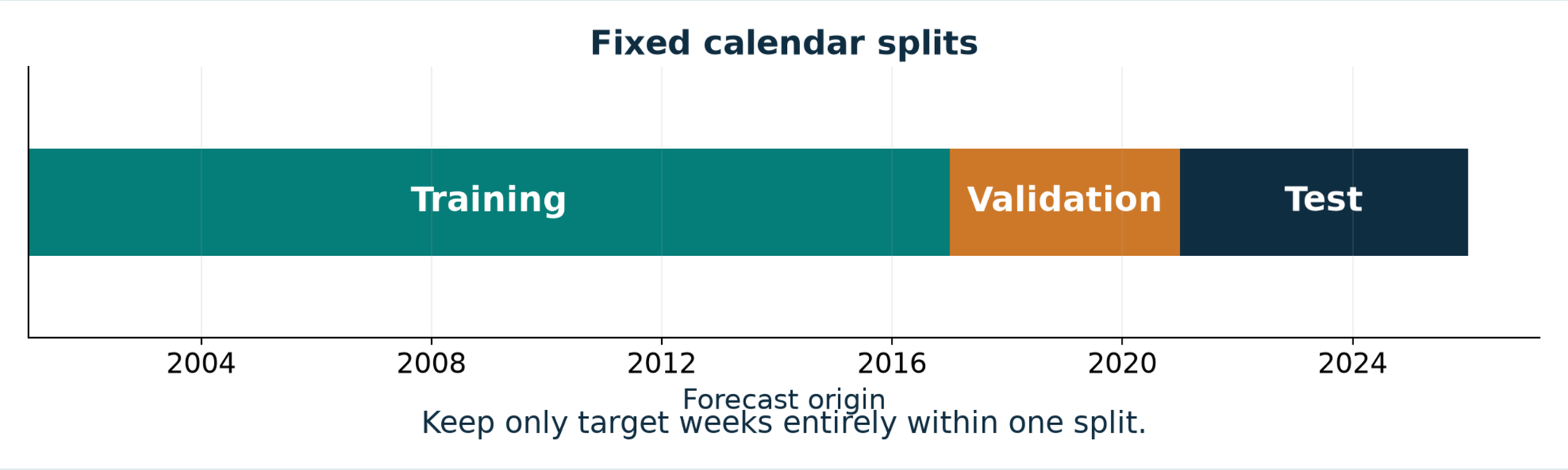
30 consecutive days (rows) × 3 daily features (columns) → one LSTM input

Illustrative day: two input events, maximum M = 5.2 → **[1.10, 0.70, 1]** before standardization.

> Input: 30 past days, M ≥ 4.5 → output: probability of at least one M ≥ 5 event in the next seven days.

The table shows values before standardization. The network uses means and standard deviations from training histories only. Features are also called channels.

Evaluate using the temporal order of the problem



Today we compare validation probabilities using Brier score, log loss and calibration.

Train 2001–2016 · Validation 2017–2020 · Test 2021–2025 was inspected previously and is excluded from this comparison.

Guided experiment: from one example to an LSTM probability

30 days × 3 catalogue channels → LSTM → weekly probability

Build one example: identify the input window, forecast origin and label.

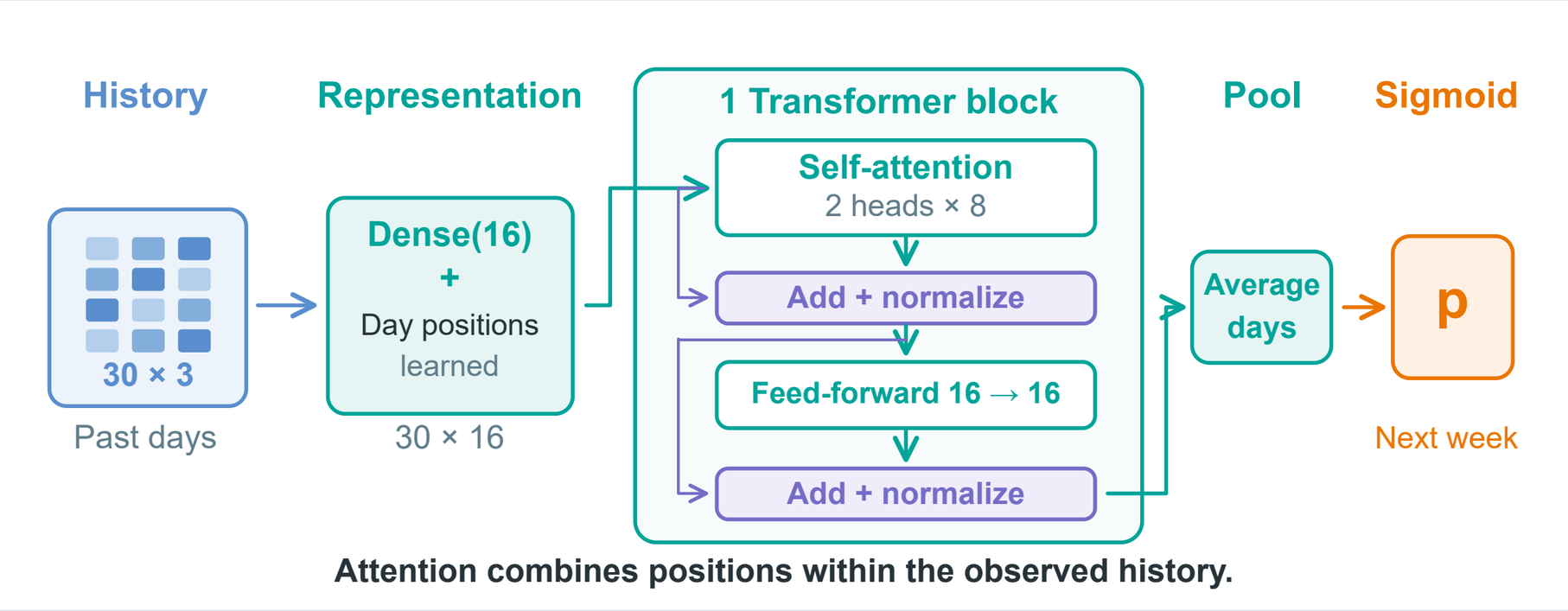
Fit references and the LSTM: read compile, fit and predict beside the model.

Evaluate: compare Brier score, log loss and calibration.

> Does the LSTM improve on training prevalence and logistic regression in this experiment?

Temporal tutorial and notebook · Seed 7 · Optional comparison: Transformer on the same history and target

Optional comparison: a Transformer on the same history



Same history and target. A different way to combine information across days.

Current temporal tutorial · 2,257 trainable parameters versus 1,297 for LSTM · One fixed, small configuration

Validation performance for the weekly occurrence task

Model	Validation Brier	Validation log loss
Training prevalence	0.188	0.563
Logistic regression	0.204	0.614
LSTM	0.182	0.548
Transformer (optional)	0.188	0.562

Lower is better. The LSTM has the lowest losses in these individual fits. Read calibration alongside them.

> These are development results from one region and one validation period.

One fit per neural model, seed 7 · 208 validation weeks · Display rounded to three decimals

What can this result tell us about the problem?

Reading the validation results

Compare each model with training prevalence.

Read Brier, log loss and calibration together.

Questions for the next experiment

Which representation and target preserve useful structure?

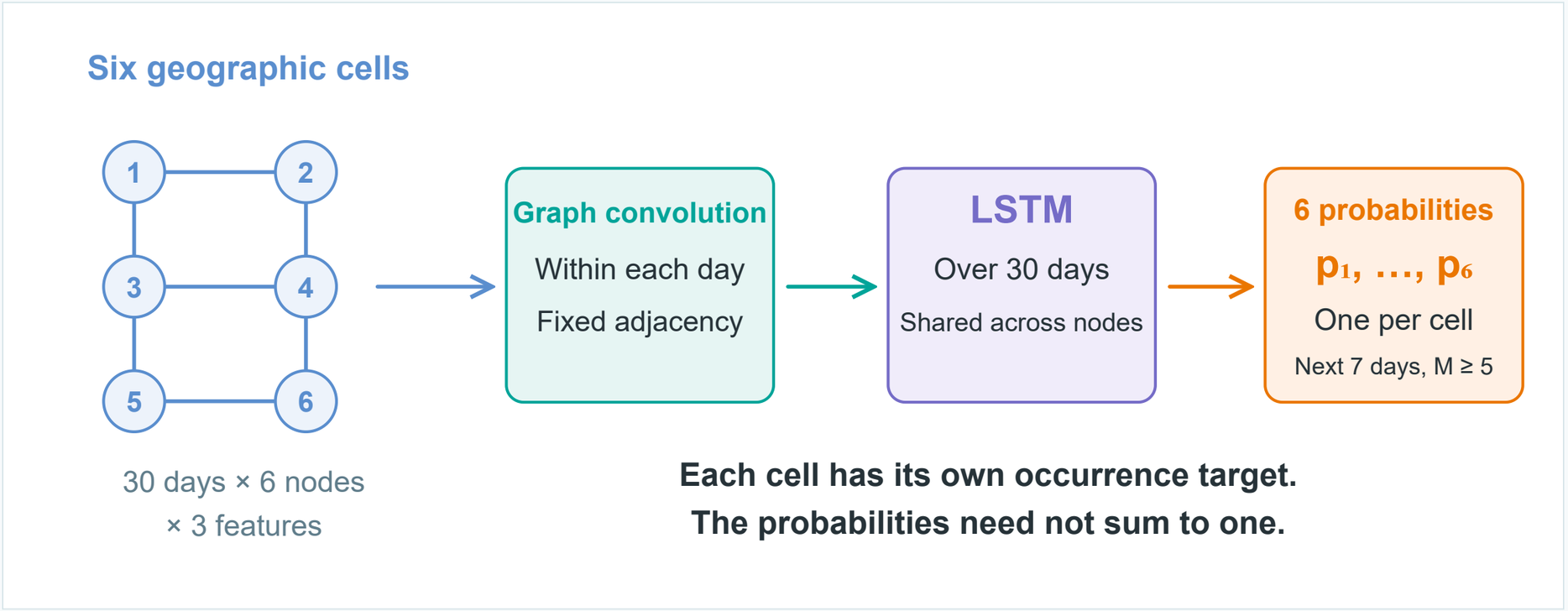
What information is available in the catalogue?

Would more data, capacity or a different training budget change the result?

> Useful deep learning depends on how we represent the phenomenon and formulate the task.

Validation also controls early stopping. One seed does not measure initialization sensitivity.

Spatial extension: Graph + LSTM



Graph connections share spatial information; the LSTM summarizes each node's history.

Input/output introduction · Complete graph example for optional exploration

What did we learn, and what would we test next?

1. What information did our daily channels preserve or discard?
2. What did the LSTM comparison with simple references establish?
3. Which research question would you explore next with maps, graphs or an activity score?

> Deep learning offers flexible tools. Their scientific value depends on the question and the evidence.

Further exploration: optional reading and experiments · LSTM references and optional extensions

Appendix · MLP: combine values at fixed lag positions

Input: 30 days × 3 channels. **Output:** one weekly occurrence probability.

```
from tensorflow import keras

inputs = keras.Input(shape=(30, 3))
fixed_lags = keras.layers.Flatten()(inputs)
hidden = keras.layers.Dense(16, activation="relu")(fixed_lags)
probability = keras.layers.Dense(1, activation="sigmoid")(hidden)
model_mlp = keras.Model(inputs, probability)
```

> Flatten gives 90 values at fixed lag positions; Dense learns nonlinear combinations.

Temporal notebook · Same catalogue channels and M ≥ 5 occurrence target over the next seven days

Appendix · CNN1D: share filters across neighbouring days

Input: 30 days × 3 channels. **Output:** one weekly occurrence probability.

```python
from tensorflow import keras

inputs = keras.Input(shape=(30, 3))
local = keras.layers.Conv1D(
    16, kernel_size=3, padding="same", activation="relu")(inputs)
combined = keras.layers.Conv1D(
    16, kernel_size=3, padding="same", activation="relu")(local)
summary = keras.layers.GlobalAveragePooling1D()(combined)
probability = keras.layers.Dense(1, activation="sigmoid")(summary)
model_cnn1d = keras.Model(inputs, probability)
```

> Conv1D scans time. A spatial CNN uses Conv2D to scan map neighbourhoods.

Temporal notebook · Width-three filters use all three channels; pooling summarizes responses across days

Appendix · the LSTM steps

Input: 30 days × 3 channels. **Output:** one weekly occurrence probability.

```
inputs = keras.Input(shape=(30, 3))
state = keras.layers.LSTM(16)(inputs)
probability = keras.layers.Dense(1, activation="sigmoid")(state)
model_lstm = keras.Model(inputs, probability)
model_lstm.compile(optimizer="adam", loss="binary_crossentropy")
history_lstm = model_lstm.fit(
    X_train, y_train, validation_data=(X_val, y_val),
    epochs=20, batch_size=64, callbacks=[stopping_lstm], shuffle=False)
p_lstm = model_lstm.predict(X_val).ravel()
brier = np.mean((p_lstm - y_val) ** 2)
```

> Define the model · compile · fit · predict · evaluate probabilities

Appendix · conditional GAN: two networks, different outputs

G: conditions + noise → sequence. **D:** sequence + conditions → real/fake score.

Generator G

```
from tensorflow import keras
from tensorflow.keras import layers
noise = keras.Input(shape=(16,), name="noise")
conditions = keras.Input(
    shape=(5,), name="conditions")
g = layers.Concatenate()([noise, conditions])
g = layers.Dense(64, activation="relu")(g)
g = layers.Dense(128, activation="relu")(g)
generated = layers.Dense(
    224, activation="tanh")(g)
generator = keras.Model(
    [noise, conditions], generated, name="G")
```

Discriminator D

```
sequence = keras.Input(
    shape=(224,), name="sequence")
d = layers.Concatenate()([sequence, conditions])
d = layers.Dense(128, activation="relu")(d)
d = layers.Dense(64, activation="relu")(d)
real_score = layers.Dense(
    1, activation="sigmoid")(d)
discriminator = keras.Model(
    [sequence, conditions], real_score, name="D")
```

> G produces 224 scaled values. D's real/fake score is not an occurrence probability.

Small, untrained teaching example · Reduced conditional-GAN architecture; no thesis replication or performance claim

Appendix · GAN training alternates two updates

Pseudocode: the update logic below is explanatory, not executable Python.

```
for each batch of observed sequences and mainshock conditions:
    draw noise; generated = G(noise, conditions)
    keep G fixed
    D_loss = BCE(1, D(observed, conditions))
           + BCE(0, D(generated, conditions))
    update only D weights to reduce D_loss

    draw fresh noise; generated = G(noise, conditions)
    keep D weights fixed, but pass gradients through D
    G_loss = BCE(1, D(generated, conditions))
    update only G weights to reduce G_loss
```

> After training: choose conditions, draw noise and use G to produce a candidate sequence.

BCE = binary cross-entropy · D learns labels real = 1 and generated = 0; G is updated toward D's real label

Appendix · quantities and scientific meaning

Quantity	Question answered
Process intensity	At what conditional rate do events occur?
Magnitude	How large is an event?
Shaking intensity	What motion or effects occur at a location?
Risk	What consequences may occur given exposure and vulnerability?

> Today’s output is a probability of a defined occurrence event.

Appendix · ETAS-inspired feature experiment

$$r_t(s) = \sum_{t-L \le t_i < s} 10^{\alpha(M_i - M_0)} \left(1 + \frac{s - t_i}{c}\right)^{-p}, \quad s \le t.$$

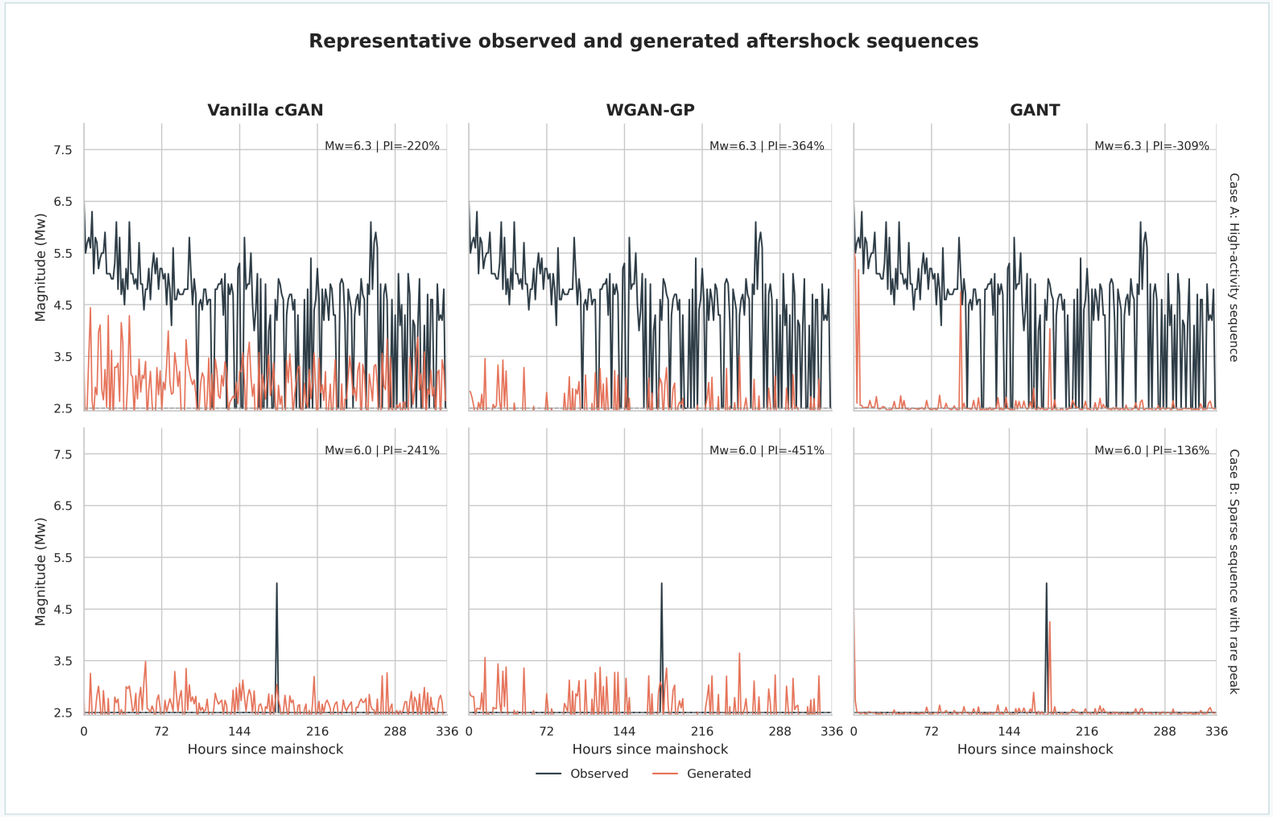
Illustrative: α=1, c=1 day, p=1.1, L=30 days.

Controlled comparison: both LSTMs use the same available history.

Representation: log1p(score) is an additional daily channel.

> Optional representation experiment. The illustrative index is not a fitted ETAS rate.

Appendix · original generative examples



2026 manuscript under review · Original PI annotations use a mean-reference denominator · Illustrative cases

Appendix · evidence and availability

Check	Meaning
Chronological, purged targets	Future target intervals do not cross partitions.
Training-only normalization	Later inputs do not set scaling parameters.
Frozen revised catalogue	Data bytes and provenance are reproducible.
Historical catalogue vintages	**Not reconstructed.**

> Reproducibility and historical information availability are separate requirements.

Appendix · sources and further reading

- **Seismic applications:** ConvNetQuake · RECAST.
- **Our published case:** Nicolis, Plaza & Salas (2021).
- **Aftershock studies:** Jara Muñoz (2025), Statistical Engineering undergraduate thesis, Universidad de Santiago de Chile; Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review.
- **Attention context and further reading:** Delgado et al. (2025) · Farfán et al. (2025).
- **Diagram style:** author's DL course, Units 2–5 · Transformer schematic matches the current temporal tutorial.
- **Materials:** practical · Further exploration · full references.